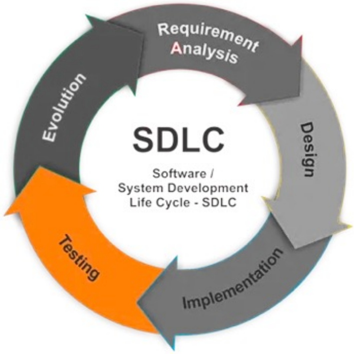
Chapter 9: Software Testing



Software Testing Life Cycle

Requirement Phase

- Identify Scope

Test Planning

- Determine Roles and Responsibilities
- Tools Selection

Test Case Design & Development

- Create Test Cases & Test Scripts
- Prepares Test Data

Test Environment Setup

- Depends on Hardware and Software

Test Execution

- Execute Test Cases with Pass/Fail Result

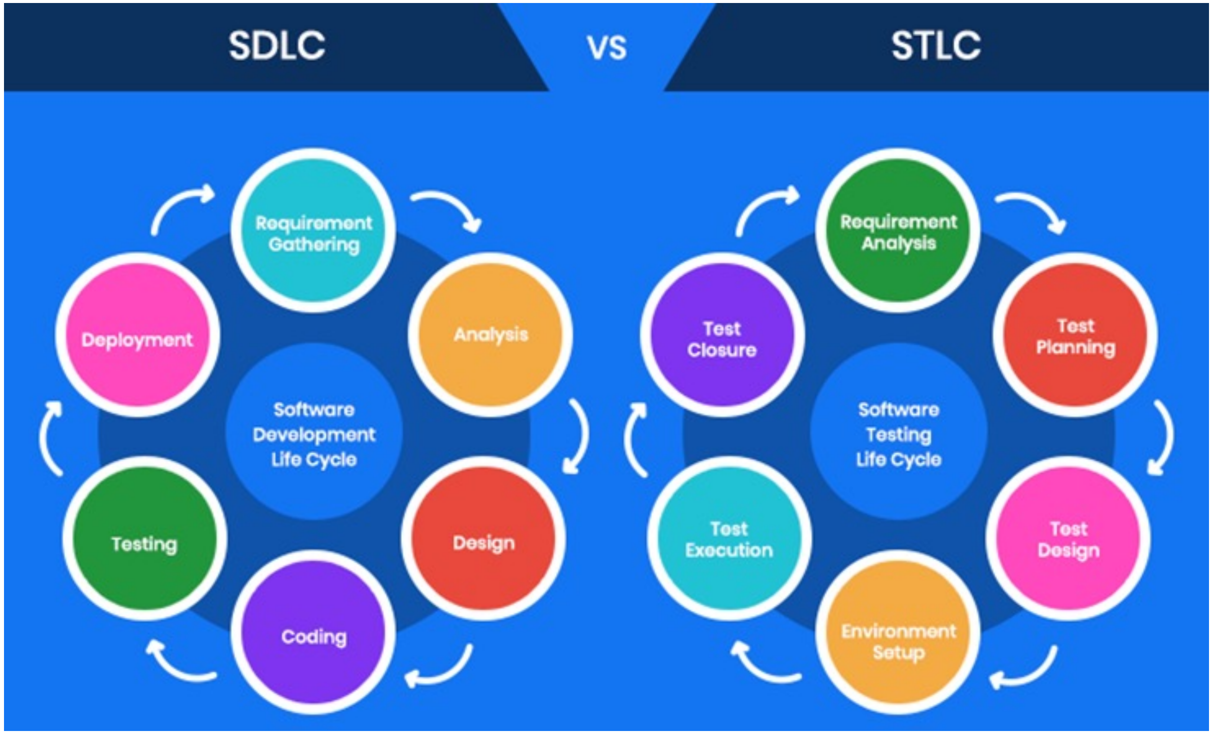
Test Cycle Closure

- Evaluate Cycle Completion Criteria

Software Testing Life Cycle

- Requirement Phase
  - Identify scope by studying and clarifying testing requirements based on user requirement
- Test Planning
  - Determine roles and responsibilities of the testers, test tools, prioritise the tasks, and monitor the test coverage
- Test Case Design and Development
  - Create test cases, automation scripts (if applicable)
  - Prepare test data (If Test Environment is available)
- Test Environment Setup
  - Decides the software and hardware conditions under which a work product is tested
  - Setup test Environment and test data
- Test Execution
  - Execute the test cases and record the test result
- Test Cycle Closure
  - Evaluate cycle completion criteria
  - A final review report on the testing to signify the end of testing is generated

SDLC vs STLC


SDLC
VS
STLC
Requirement Gathering
Analysis
Design
Coding
Testing
Deployment
Software Development Life Cycle
Requirement Analysis
Test Planning
Test Design
Environment Setup
Test Execution
Test Closure
Software Testing Life Cycle

SDLC vs STLC

SDLC is responsible for

- collecting requirements
- creating features accordingly

STLC is responsible for

- creating tests for the collected requirements
- verifying that features meet those requirements

Testing Artifacts

Documents Required Before Testing

Before Testing

- Test plan doc
- Test cases doc
- Test design specs

Testing Artifacts (Before Testing)

- **Test Plan** outlines the test strategy, testing objectives, resources required for testing, test schedule and test deliverables.
- **Test Cases** will examine the expected and actual result and gives a pass or fail status.
  - Usually contain test ID, description, test procedure, test data, expected result, actual result, pass/fail status.
- **Test Design** specification refines the test approach and identifies the features to be covered by the design and its associated tests.

How to write a TEST CASE?

https://youtu.be/BBmA5Qp6Ghk

Test Case ID	Test Case Description	Test Steps	Test Data	Expected Results	Actual Results	Pass/Fail
TU01	Check Customer Login with valid Data	1.Go to site http://demo.guru99.com 2.Enter UserId 3.Enter Password 4.Click Submit	Userid = guru99 Password = pass99	User should Login into an application	As Expected	Pass
TU02	Check Customer Login with invalid Data	1.Go to site http://demo.guru99.com 2.Enter UserId 3.Enter Password 4.Click Submit	Userid = guru99 Password = glass99	User should not Login into an application	As Expected	Pass

Test Scripts

- What is Automated Testing
  - https://youtu.be/Nd31XiSGJLw

- **Test Scripts** - A set of instructions to test an application automatically.
- **Test Data** – Use relevant historical data from the client or create meaningful contextual data for testing.
- **Test Data and Test Scripts** are required for automated testing during test execution phase

Ways to create test script are

1. Record/playback
2. Keyword/data-driven scripting
3. Writing Code Using Programming Language

Record/playback:

The Easiest Way To Start Automating Tests

1. Use tool to capture the actions and automatically turn them into a test script.
2. Then "playback" or rerun the test steps to make sure they can run like it's supposed to.

Can become stressful when the UI of the application changes frequently. In these cases, the recorded tests might easily become broken.

Demo of Record And Playback Testing Tools **Katalon**

https://academy.katalon.com/courses/record-playback-testing/?utm_source=sth&utm_medium=blog&utm_campaign=record_and_playback

Keyword/data-driven scripting

A scripting technique that uses data files which contain keywords related to the application being tested.

A set of keywords are associated with actions (or functions).

Keywords	Description
Login	Login to website
Email	Send Email
Logout	Log out from website
Notification	Find unread notifications
Openbrowser	Opening of browser
Typtext	keystrokes
Lclick	Mouse left click

Example of a Test Script

For example, to check the login function on a website, your test script might do the following:

1. Specify how the automation tool can locate the “Username” and “Password” fields in the login screen. Let us say, by their CSS element IDs.
2. Load the website homepage, then click on the “login” link. Verify that the Login screen that appears and the “Username” and “Password” fields are visible.
3. Next, type the username “Charles” and password “123456” identify the “Confirm” button and click it.
4. They need to specify how a user can locate the title of the Welcome screen that appears after login- say, by its CSS element ID.
5. Verify that the title of the Welcome screen is visible.
6. Read the title of the welcome screen.
7. Insert that the title text is “Welcome Charles”.
8. If the title text is as per the expectation, a record that the test passed. Otherwise, the test failed.

Difference Between Test Case And Test Script

Test Case	Test Script
Test case is a step by step procedure that is used to test an application.	Test script is a set of instructions to test an application automatically.
Test cases are used for manual testing environment.	Test script is used in the automation testing environment.
It is done manually.	It is done according to the scripting format.
The test case template contain test ID, description, test procedure, test data, expected result, actual result, pass/fail status, etc.	In the Test Script, we can use different commands to develop a script.

User Acceptance Testing

What is UAT?	User Acceptance Testing (UAT), also known as end-user testing, is defined as testing the software by the user or client to determine whether it can be accepted or not.
When Is It Performed?	This is typically the last step before the product goes live or before the delivery of the product is accepted.
Who Performs UAT?	Users or client – someone who is buying a product or who has had a software custom-built through a software service provider or the end-user
Key activities of each UAT phase	UAT Test Initiation, UAT Test Design, UAT Test Execution, UAT Test Closure

Summary

- Software Testing Lifecycle – **Requirement Phrase**, **Test Planning**, **Test Case Design and Development**, **Test Environment Setup**, **Test Execution** and **Test Cycle Closure**.
  - **Test Plan** outlines the test strategy, testing objectives, resources required for testing, test schedule and test deliverables.
  - **Test Cases** examine the expected and actual result and gives a pass or fail status.
    - The test case template contains test ID, description, test procedure, test data, expected result, actual result, pass/fail status, etc.

Summary

- **Test Data** – Use relevant historical data from the client or create meaning contextual data for testing.
- **Test Scripts** - A set of instructions to test an application automatically.
- **Test Data and Test Scripts** are required for automated testing during testing
- **User Acceptance Testing (UAT),** also known as end-user testing, is defined as testing the software by the user or client to determine whether it can be accepted or not. Upon acceptance, the product is ready to go ‘live’.

References

Testing framework:

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- https://www.softwaretestingmaterial.com/tdd-vs-bdd-vs-atdd

Testing type, technique & Tactics:

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- https://www.janio.careers/post/grey-box-testing-with-a-janio-qa-engineer

Measurement in software testing - https://www.vskills.in/certification/tutorial/measurement-in-software-testing/

Related Process

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- https://www.panaya.com/blog/testing/what-is-uat-testing
- https://qatestlab.com/resources/knowledge-center/5-types-of-test-data-that-should-be-used-during-software-testing/